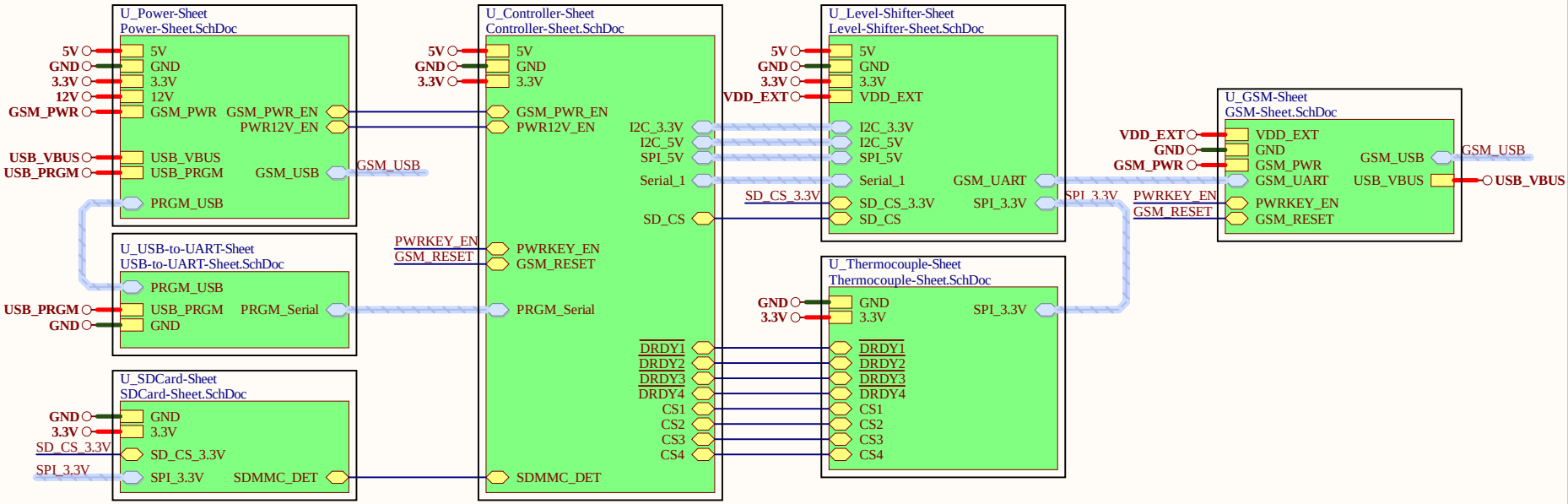
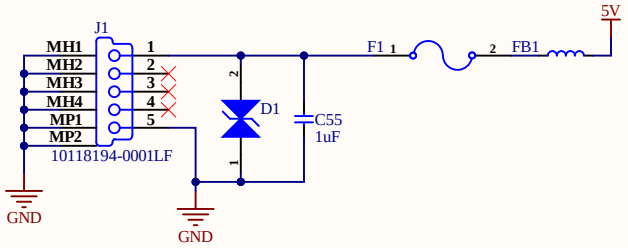


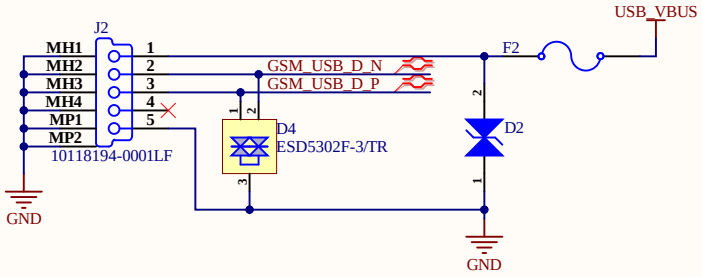
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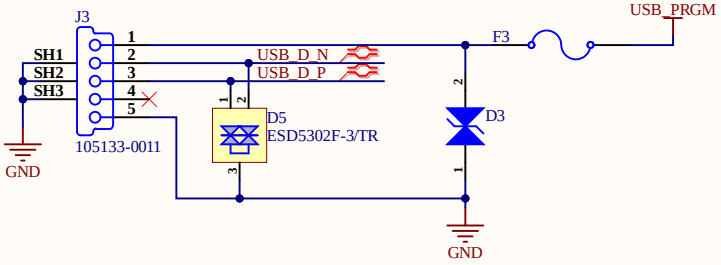
USB FOR BOARD POWER



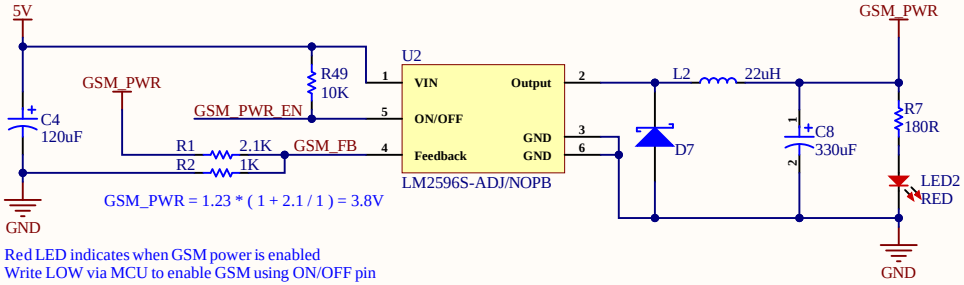
USB FOR GSM UPDATES



USB PROGRAMMING MEGA2560

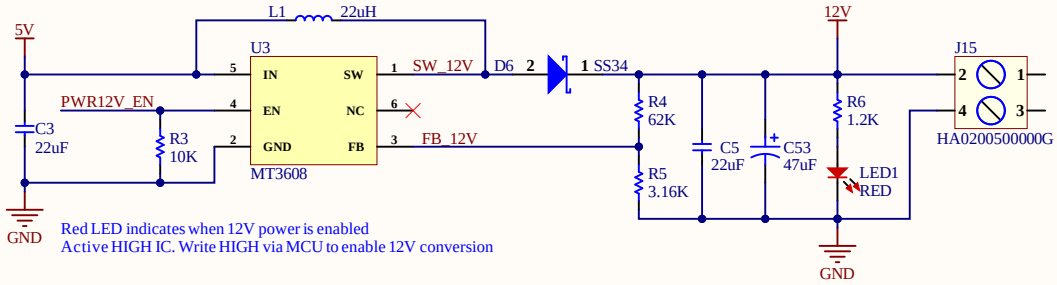


BUCK FOR GSM POWER 5V TO 3.8V



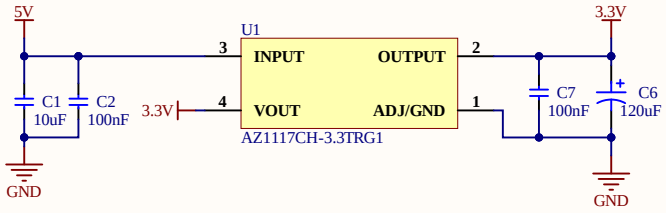
Red LED indicates when GSM power is enabled
Write LOW via MCU to enable GSM using ON/OFF pin

BOOST FOR WIKA SENSOR 5V TO 12V

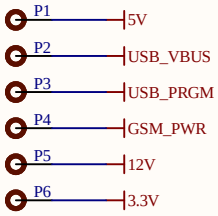


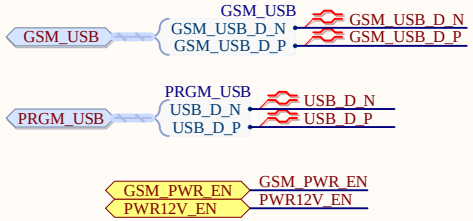
Red LED indicates when 12V power is enabled
Active HIGH IC. Write HIGH via MCU to enable 12V conversion

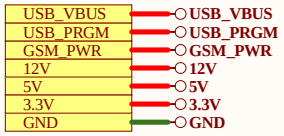
LDO 5V TO 3.3V



Test Points



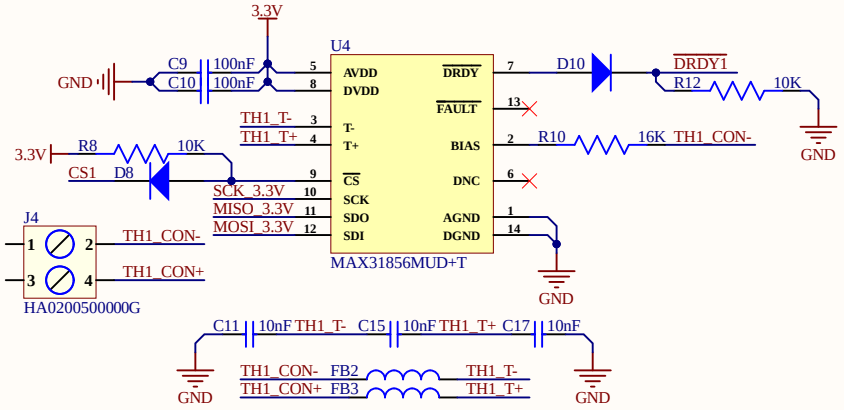




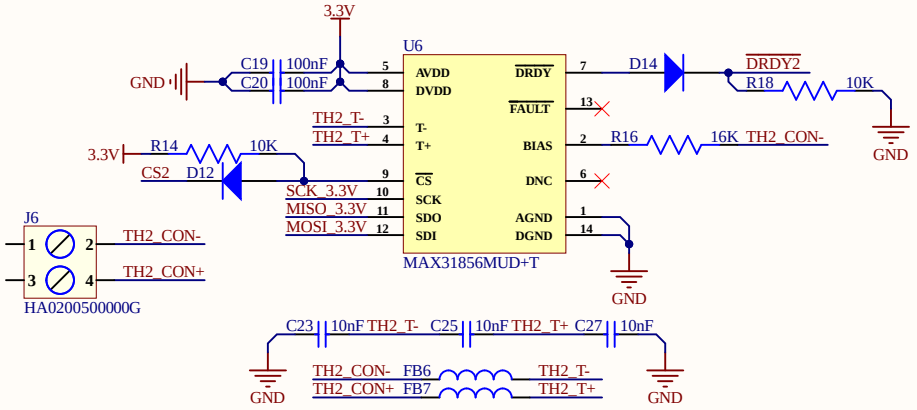
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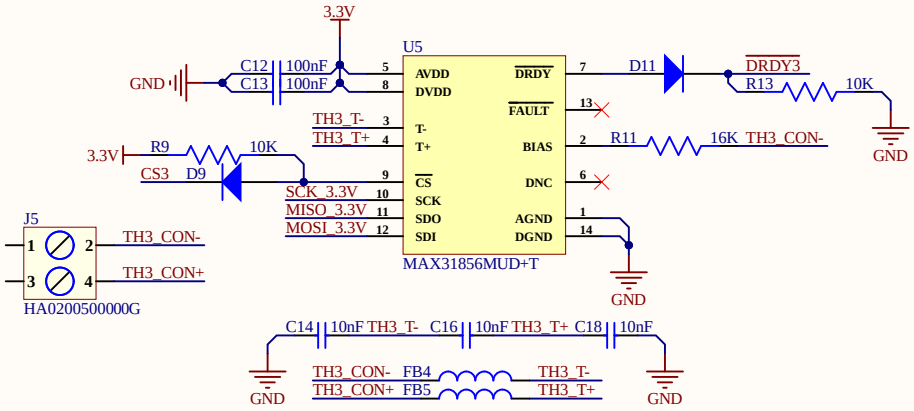
Thermocouple Input - 1



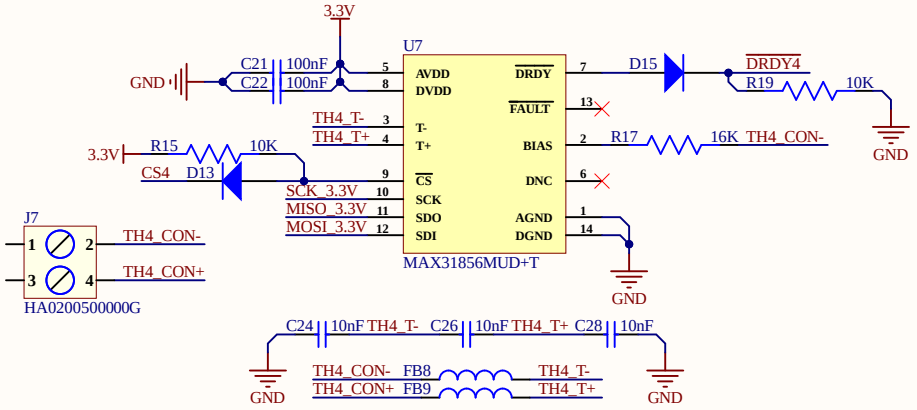
Thermocouple Input - 2



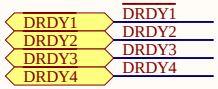
Thermocouple Input - 3

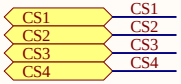


Thermocouple Input - 4







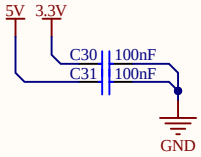


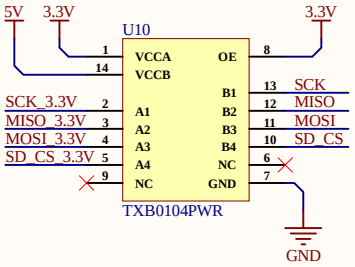


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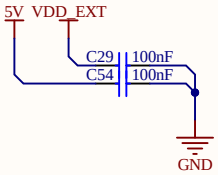
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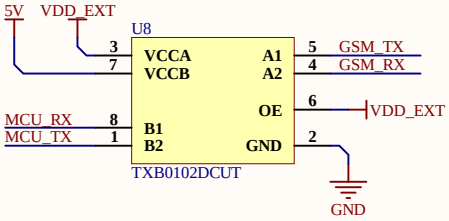
Level Shifter for SPI



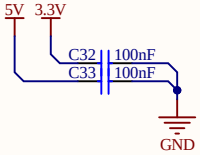


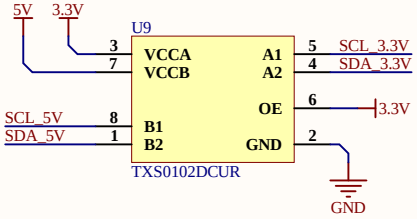
Level Shifter for Serial

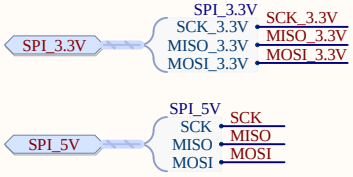


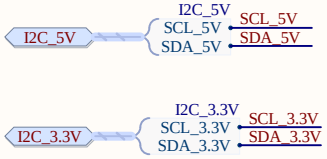


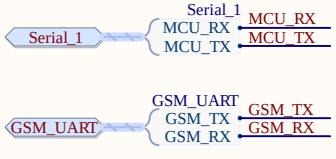
Level Shifter for I2C

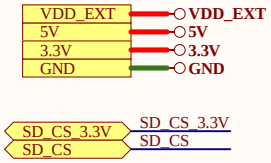






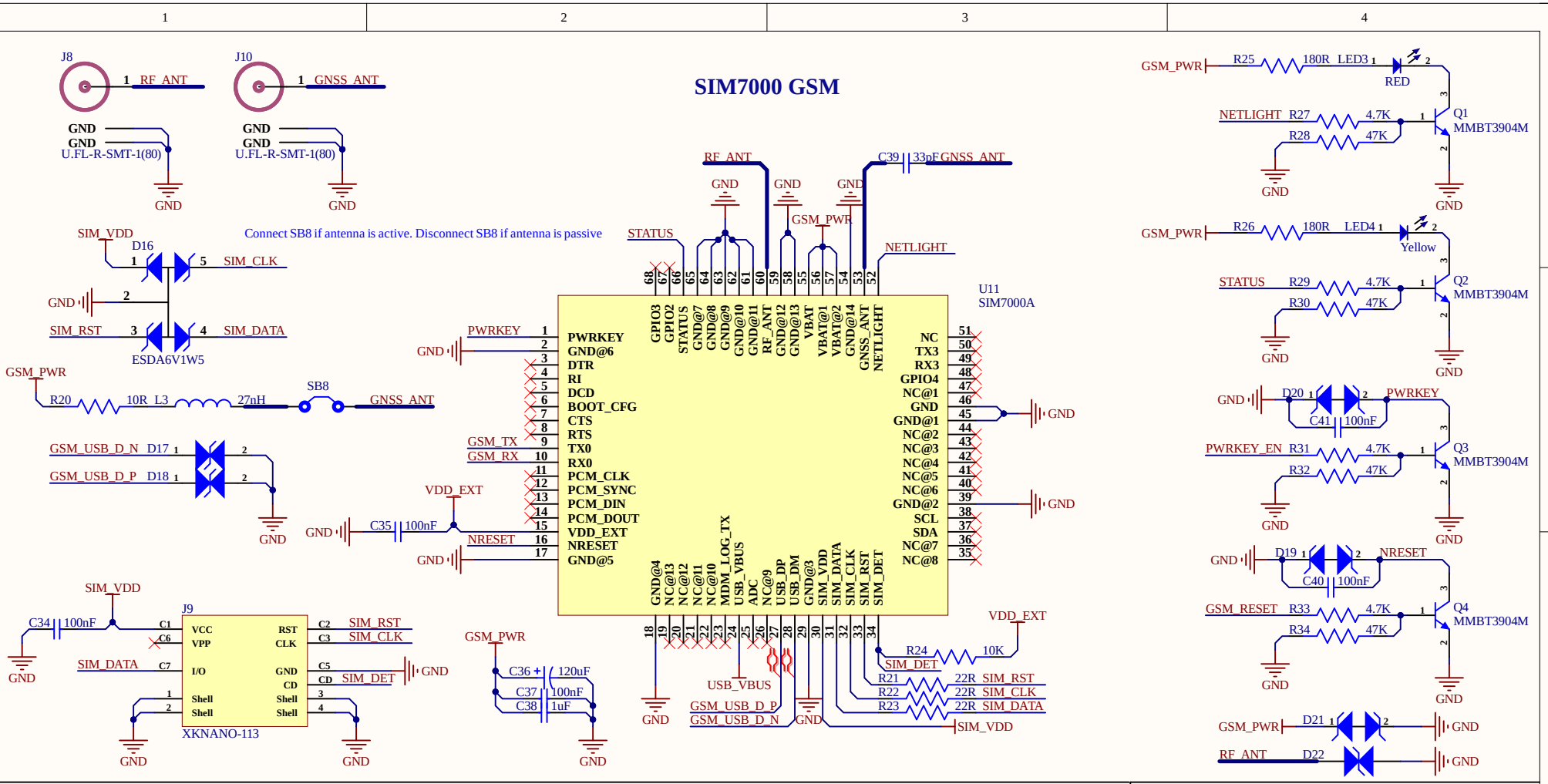


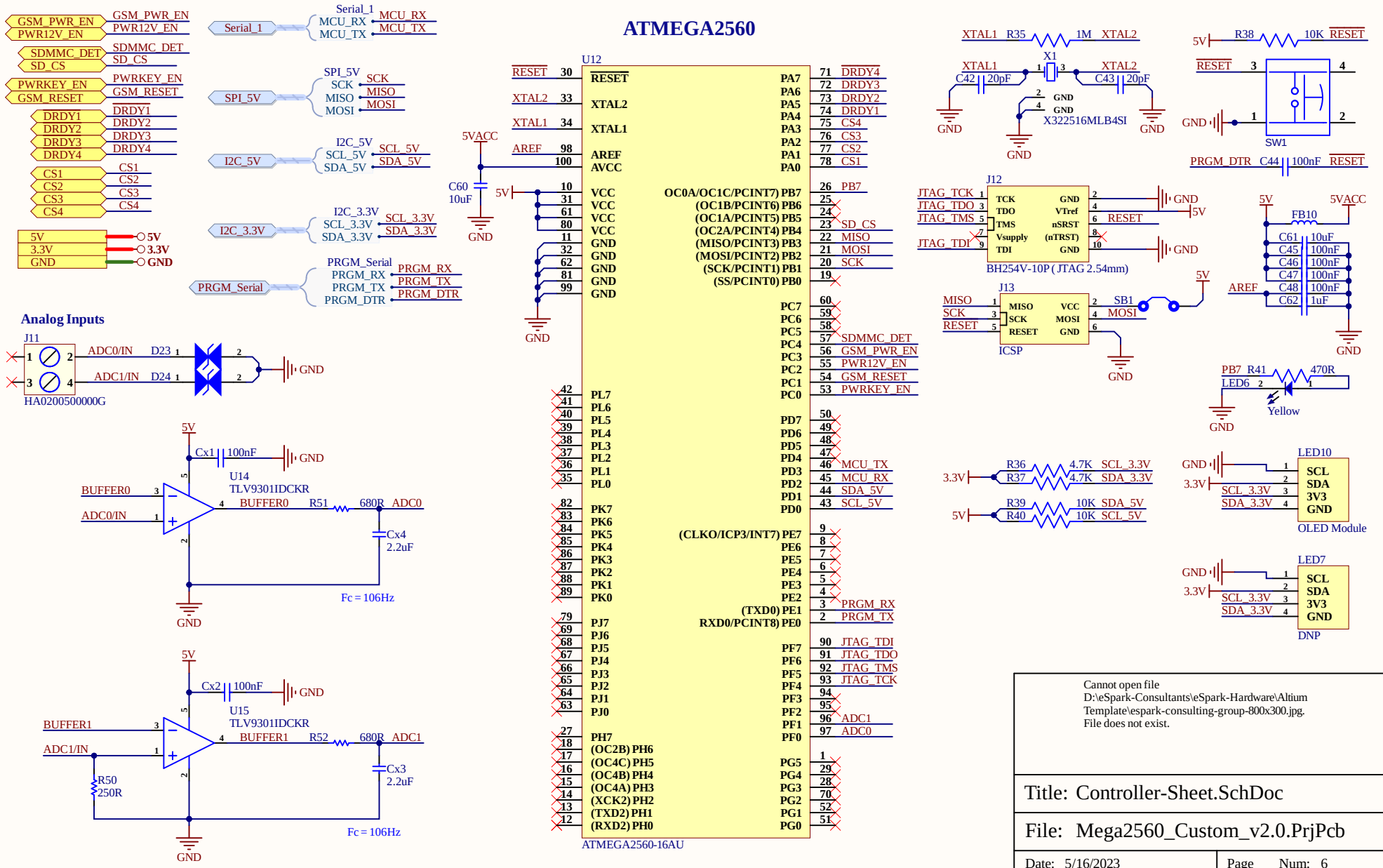




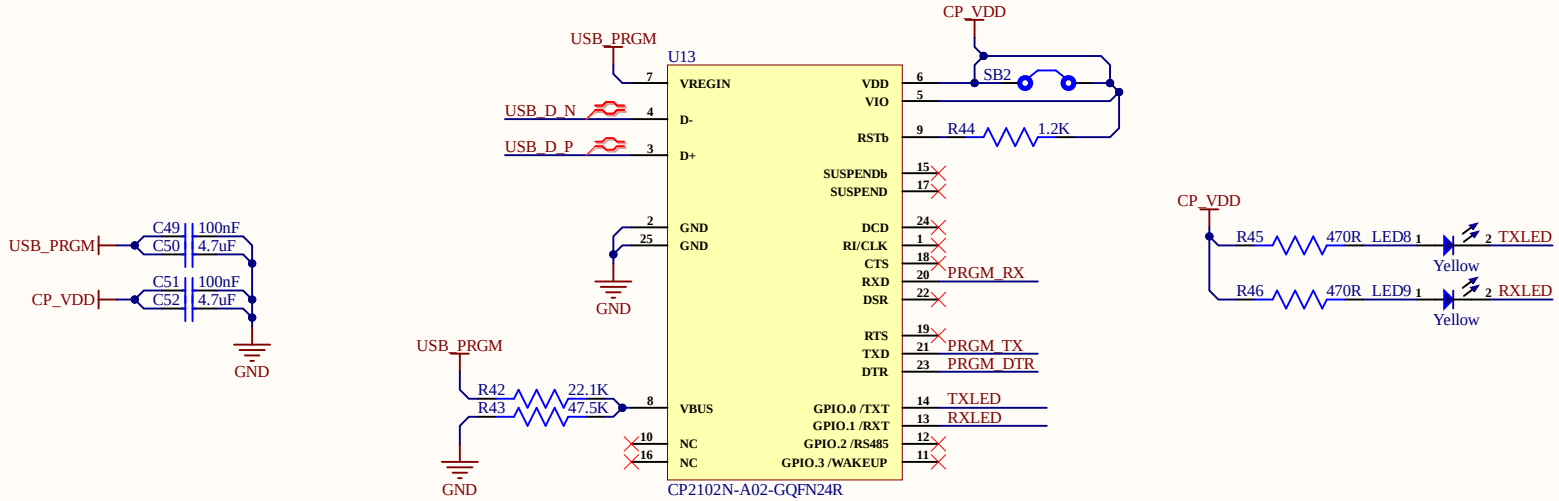
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Drawn By: Huzaifa Ali	Page	Total: 8


SIM7000 GSM
J8
J10
RF_ANT
GNSS_ANT
GND
U.FL-R-SMT-1(80)
SIM_VDD
D16
SIM_CLK
SIM_RST
SIM_DATA
ESDA6V1W5
Connect SB8 if antenna is active. Disconnect SB8 if antenna is passive
GSM_PWR
R20
10R
L3
27nH
SB8
GSM_USB_D_N
GSM_USB_D_P
D17
D18
C35
100nF
VDD_EXT
PWRKEY
NRESET
GSM_TX
GSM_RX
STATUS
NETLIGHT
C39
33pF
U11
SIM7000A
PWRKEY
GND@6
DTR
RI
DCD
BOOT_CFG
CTS
RTS
TX0
RX0
PCM_CLK
PCM_SYNC
PCM_DIN
PCM_DOUT
VDD_EXT
NRESET
GND@5
GPIO3
GPIO2
STATUS
GND@7
GND@8
GND@9
GND@10
GND@11
RF_ANT
GND@12
GND@13
VBAT
VBAT@1
VBAT@2
GND@14
GNSS_ANT
NETLIGHT
NC
TX3
RX3
GPIO4
NC@1
GND
GND@1
NC@2
NC@3
NC@4
NC@5
NC@6
GND@2
SCL
SDA
NC@7
NC@8
GND@4
NC@13
NC@12
NC@11
NC@10
MDM_LOG_TX
USB_VBUS
ADC
NC@9
USB_DP
USB_DM
GND@3
SIM_VDD
SIM_DATA
SIM_CLK
SIM_RST
SIM_DET
J9
XKNANO-113
VCC
VPP
I/O
Shell
RST
CLK
GND
CD
C34
C36
120uF
C37
C38
1uF
R24
10K
R21
R22
R23
22R
SIM_RST
SIM_CLK
SIM_DATA
SIM_DET
USB_VBUS
GSM_USB_D_P
GSM_USB_D_N
R25
180R
LED3
RED
NETLIGHT
R27
4.7K
R28
47K
Q1
MMBT3904M
R26
LED4
Yellow
R29
R30
Q2
D20
C41
100nF
PWRKEY_EN
R31
R32
Q3
D19
C40
NRESET
GSM_RESET
R33
R34
Q4
D21
RF_ANT
D22
GSM_UART
GSM_USB
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GSM_RESET
VDD_EXT
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Date: 5/16/2023
Page Num: 5
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ATMEGA2560
Analog Inputs
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Page Num: 6
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Page Total: 8

USB-to-UART for Programming





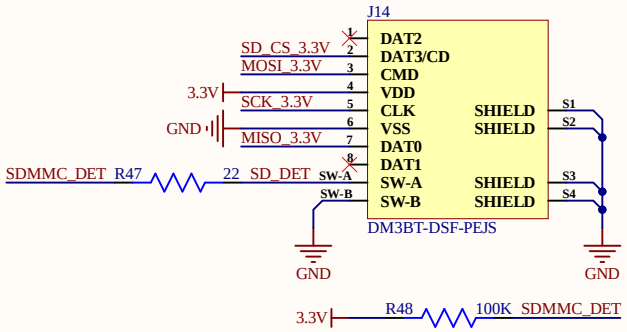


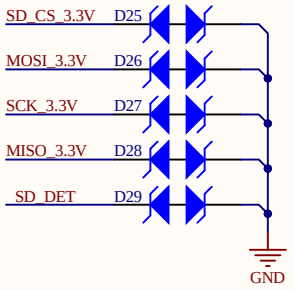


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SD Card Interface











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